

Nandini Mehta

SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Data and enterprise software engineer with **4 years of hands-on development experience** across **Python data pipelines, relational databases, cloud backend services and business application integrations**. Strong in **Python, object-oriented programming, SQL, PostgreSQL, ETL/ELT, REST APIs and data validation**. Delivered a repeatable workflow for more than **25 GB of engineering time-series data** and production cloud services processing over **50,000 daily events**. Experienced in translating technical and business requirements into modular solutions, schema changes, testable workflows and documented releases. Hands-on with **Google Cloud, Docker, Kubernetes, Git and CI/CD** with transferable foundations for Azure, AWS and enterprise application environments.

TECHNICAL SKILLS

Data Engineering and ETL: ETL/ELT Workflows | Data Ingestion | Data Migration | Data Transformation | Data Cleansing | Data Validation | Batch Processing | Time-Series Processing | Data Quality

Python and Software Engineering: Python | Object-Oriented Programming | pandas | NumPy | Django | Flask | Java | Go | PHP | JavaScript | Modular Architecture

Databases and Integration: SQL | PostgreSQL | Oracle SQL | Relational Schema Design | Database Migrations | REST APIs | Third-Party System Integration | Structured Reporting

Cloud and Data Platforms: Google Cloud | Cloud-Native Services | Databricks Concepts | Azure Data Platform Concepts | AWS Data Platform Concepts | Event-Driven Workflows

DevOps and Delivery: Git | CI/CD | Docker | Kubernetes | Linux | Elasticsearch | Kibana | Unit Testing | Integration Testing | Deployment Support

Enterprise Solutions: Business Workflow Automation | Requirements Analysis | Technical Solution Design | Configurable Business Applications | Testing and Release Support

Collaboration: JIRA | Confluence | Agile Delivery | Technical Documentation | Stakeholder Communication | International Teamwork

WORK EXPERIENCE

OBM Pinpointing with Machine Learning (Master Thesis)

Apr 2025 to Dec 2025, Grade 1.9

IAV GmbH, Berlin, Germany

- Engineered a modular **Python data-processing pipeline** to ingest, transform and validate more than **25 GB of real OBD and engine time-series data** for emission-anomaly analysis.
- Implemented reusable stages for **data cleaning, feature selection, robust scaling, dimensionality reduction and anomaly detection** with traceable intermediate results.
- Created **automated quality checks, reproducible workflows and technical documentation** while collaborating with engineering and research stakeholders.

Software Developer

Dec 2023 to May 2024

ChargeHorizons GmbH, Buxtehude, Germany

- Developed production **backend and data services in Python and Go** with PostgreSQL and Google Cloud processing more than **50,000 daily events**.
- Designed **relational schemas, data ingestion flows and REST API integrations** connecting internal systems with external platforms.
- Built **CI/CD pipelines, Docker containers and Kubernetes deployments** plus Elasticsearch and Kibana monitoring for data quality, schema drift and troubleshooting.

Junior Python Developer

Oct 2021 to Mar 2023

Intellial Solutions Pvt. Ltd., Ahmedabad, India

- Developed **modular Python applications and automated data pipelines** that reduced QA cycle time by **25 percent**.
- Built and maintained more than **five Django and Flask applications** with PostgreSQL serving around **2,000 daily users**.
- Implemented **database migrations, REST APIs, validation logic and reusable processing components** for reliable transformation and reporting.

ADDITIONAL EXPERIENCE

Junior PHP Developer

Jun 2020 to Oct 2021

World4Business, Bhavnagar, India

- Built backend and full-stack business applications with **Laravel, Symfony, React.js, Node.js and relational databases** in a six-person agile team.
- Supported **requirements clarification, workflow changes, third-party integrations, testing and deployment** using Git, JIRA and Confluence.

Java Developer Intern

Dec 2017 to Jun 2018

Chanakya Engineering Institute, Bhavnagar, India

- Developed and tested **Java, PHP and Oracle SQL application features** while supporting database handling, debugging and defect resolution.

SELECTED PROJECTS

Document Classification and Structured Extraction

Personal Project | Python | NLP | Hugging Face Transformers

- Built a modular **document ETL pipeline** for OCR, classification, information extraction and conversion of unstructured files into validated structured records.
- Separated ingestion, transformation and validation into **reusable production-style workflow stages** for downstream analytics and semantic applications.

RAG-Based LLM Data Pipeline

Personal Project | Python | Vector Database | Retrieval-Augmented Generation

- Implemented **document indexing, chunking, metadata processing and vector retrieval** for scientific and technical content.
- Designed **traceable ingestion and retrieval workflows** with source grounding, structured outputs and validation checkpoints.

Manufacturing Data Integration and Semantic Search

Personal Project | Python | RDF | OWL | SPARQL | QUDT

- Integrated **production assets, process steps, sensor observations and units** into a structured manufacturing knowledge graph.
- Implemented **data mapping, SPARQL queries and semantic search** to connect heterogeneous technical sources and prepare machine-learning-ready datasets.

Workflow Automation with LangGraph

Personal Project | Python | LangGraph | Tool Integration

- Built a **multi-step workflow application** that coordinates tools, manages state and returns validated structured outputs.
- Automated **repeatable task execution and result validation** for research, document-processing and enterprise workflow use cases.

EDUCATION

M.Sc. Automotive Software Engineering, TU Chemnitz, Germany

Apr 2022 to Mar 2026, Grade 2.4

- Coursework in **artificial intelligence, databases, networks, distributed systems and software engineering**. Master thesis completed with IAV GmbH on large-scale OBD time-series processing and machine learning.

B.E. Information Technology, Shantilal Shah Engineering College, India

2018 to 2021, Grade 1.4

- Coursework in machine learning, Internet of Things, big data analytics, data structures, REST APIs and databases.
- **Bachelor Thesis (Wildlife Protecting System)**: An embedded railway-track safety system that detects wild animals using PIR sensors and sends alert signals to the nearest station. Equipment: Arduino, PIR sensor, LED and buzzer.

Diploma in Information Technology, Sir Bhavsinhji Polytechnic Institute, India

2015 to 2018, Grade 1.3

- Technical foundation in programming, databases, operating systems, network security and software development.

LANGUAGES

English (Fluent) | German (B1, actively improving) | Hindi (Native) | Gujarati (Native)

REFERENCES

Dipl. Ing. Marco Moser | Supervisor, Smart Data Analytics OBD, IAV GmbH | marco.moser@iav.de
Jeroen Trzaska | CEO, ChargeHorizons GmbH | jeroen@chargehorizons.com